

The Physical and Contextual Limits of Quantum Speedup

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Quantum computation is not many classical computations running in parallel, but controlled interference in Hilbert space. Amplitude is spread, problem structure is folded into phases, and measurement detects the invariant made visible by recombination. What is exposed is usually not a branchwise table of values, but a structure-matched statistic such as a parity, period, symmetry class, or marked subspace. Quantum advantage can arise when the relevant structure is matched to an end-to-end efficient interference pattern—including state preparation, oracle access, unitary implementation, and readout—under the constraints of measurement, thermodynamic irreversibility, and contextuality.

Keywords: quantum computation, quantum interference, structure-matched recombination, contextuality, quantum speedup

I. INTRODUCTION

A common description of quantum computation says that a quantum computer evaluates a function on many inputs “at once.” The image is seductive: a register is placed in a superposition, a function is applied to all components, and a final measurement extracts the desired answer. This picture is often invoked when discussing Deutsch–Jozsa [1], Simon’s algorithm [2], Shor’s algorithm [3], Grover search [4], and related oracle or Fourier-sampling procedures [5, 6].

Taken literally, however, this picture is wrong. A quantum computer does not return the list of all computed values. The components of a superposition cannot be individually inspected, copied, deleted, or archived as classical alternatives. They are amplitudes in a Hilbert-space vector, and only suitably arranged interference effects become accessible in a final measurement context, where a context is understood as a compatible set of observables, or equivalently a choice of measurement basis in the non-degenerate finite-dimensional case.

The thesis of this article is the following: Quantum speedups are not many classical computations performed in parallel. They arise when the algebraic or spectral structure relevant to a task can be embedded into coherent amplitudes and then recombined so that the desired invariant is concentrated in a chosen measurement context. Coarse-grained outputs, such as partition labels, are common because invariants are natural targets for interference, but coarseness is not itself the source of the advantage. At the same time, copying and erasure are physically constrained by no-cloning, no-deletion, and Landauer irreversibility, while contextuality, when a Kochen–Specker scenario is present, obstructs a single context-independent value assignment across incompatible measurement contexts.

A related ambiguity concerns the word “universal.” Circuit or unitary universality means that, for each fixed

finite register size, a finite gate set can approximate the relevant unitary transformations. Turing universality, by contrast, concerns effective procedures on inputs of unbounded length. These are not the same notion. A family of quantum circuits defines an algorithm in the complexity-theoretic sense only when an effective uniform rule generates the appropriate circuit for each input size. A single finite circuit, or even a universal finite gate set, is not by itself a model of partial recursive functions.

This formulation is deliberately restrictive. It does not deny quantum speedup. It denies that branchwise classical computation explains it, and it denies that circuit universality should be conflated with Turing universality.

The claim is not meant to settle interpretive questions about Everettian or many-worlds accounts of quantum computation. Advocates of such accounts have argued that quantum algorithms can be understood in terms of computations distributed across branches or worlds [7–9]. The narrower point made here is operational and model-theoretic: the components of a superposition are not independently readable classical records, and the computational advantage is realized only when the global unitary evolution and final measurement context convert phase relations into an accessible, structure-matched statistic.

The paper is therefore mainly clarificatory. Its technical contribution is the leakage construction in Appendix A, which gives a simple unitary interpolation between the Bernstein–Vazirani interference pattern and ordinary value-query behavior by exporting a tunable amount of which-path information to an inaccessible witness.

II. SUPERPOSITION IS NOT BRANCHWISE READOUT

Let an n -qubit state be written in the computational basis as

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle.$$

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The dimension of the Hilbert space is 2^n , and this exponential dimension is often used to motivate quantum parallelism. But the amplitudes α_x are not classical memory cells. A measurement yields one outcome per run, with probabilities determined by the Born rule. Recovering the amplitudes requires many similarly prepared systems and many measurement settings, that is, tomography.

Thus a superposition can be acted on linearly, but it cannot be read branch by branch. Holevo's theorem gives one precise expression of this limitation. For an ensemble of states on a Hilbert space $\mathcal{H}$, the accessible classical information is bounded above by the Holevo quantity χ , and $\chi \leq \log_2 \dim \mathcal{H}$. Hence, for an unassisted ensemble encoded in an n -qubit system, the accessible classical information per use is at most n bits [6, 10]. (Entanglement-assisted communication is a different resource setting.)

The state vector may contain exponentially many amplitudes, but the accessible classical readout is sharply constrained. This is the first reason why the naive parallelism metaphor is misleading. The superposed components can interfere, but they are not simultaneously available classical facts.

III. THE USEFUL MECHANISM: STRUCTURE-MATCHED RECOMBINATION

Many oracle-based and Fourier-sampling quantum algorithms have the schematic form

$$\begin{aligned} |0\rangle^{\otimes n} &\xrightarrow{\text{spread}} \frac{1}{\sqrt{2^n}} \sum_x |x\rangle \\ &\xrightarrow{\text{compute}} \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \\ &\xrightarrow{\text{detect}} \text{measurable interference pattern.} \end{aligned}$$

The middle step may be produced by phase kickback. If

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

and the target register $|y\rangle$ is prepared in $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, then

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

The values of $f(x)$ are not read out individually. They are converted into phases. The final measurement is useful only if those phases are recombined in a basis matched to the structure one wants to detect.

This is the essential mechanism sometimes summarized as spread, compute, detect, or spread, fold, detect. The algorithm spreads amplitude, encodes structure coherently, and then folds the amplitudes back together so that constructive and destructive interference concentrate a relevant invariant in a chosen measurement context.

Here “structure-matched” is meant operationally, but not vacuously. In the examples below the match has two

main forms: character, period, or hidden-subgroup structure, as in Bernstein–Vazirani, Simon, and Shor, where a Fourier transform turns an invariant into spectral concentration; and spectral or geometric concentration in a low-dimensional invariant subspace, as in Grover search and the Hein–Tanner lattice examples.

Deutsch–Jozsa is the simplest example. After phase kickback, the query register is in the state

$$\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle.$$

Applying the final Hadamard transform $H^{\otimes n}$ gives the amplitude of $|0^n\rangle$ as

$$\langle 0^n | H^{\otimes n} \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \right) = \frac{1}{2^n} \sum_x (-1)^{f(x)}.$$

This amplitude is ± 1 for constant functions and 0 for balanced functions. The promise makes the global predicate visible as complete constructive or destructive interference. The Deutsch–Jozsa promise problem therefore gives an exact one-query separation from deterministic classical black-box query algorithms. It is used here as a transparent interference example, not as a claim of an exponential bounded-error separation: randomized classical sampling distinguishes constant from balanced with $O(1)$ queries at bounded error.

This example also clarifies what the one-query quantum result concerns. It is not the recovery of a unique classical table of values. It is an exact solution of a decision problem whose answer is a coarse-grained class, because that class is exactly the invariant concentrated by the final interference pattern. For the original one-bit Deutsch problem, the four functions

$$f : \{0, 1\} \rightarrow \{0, 1\}$$

are not individually identified. They are partitioned into the two classes

$$\Pi_D = \{\{00, 11\}, \{01, 10\}\},$$

where the strings denote the ordered value pairs $f(0)f(1)$. Equivalently, the accessible answer is the parity

$$f(0) \oplus f(1),$$

or, in the usual terminology, the distinction between constant and balanced functions. The final interference pattern is therefore not a readout of $f(0)$ and $f(1)$ separately. It is a readout of the partition cell containing f .

This is one natural decision-theoretic form of the mechanism. A problem instance set Ω is supplied with a partition Π , and the task is not to determine $\omega \in \Omega$ but to determine its equivalence class $[\omega]_\Pi$. Quantum speedup is possible only when the oracle and the measurement basis can be arranged so that different cells of Π are represented by distinguishable interference patterns, while distinctions inside a cell are deliberately discarded. Earlier

work on quantum state identification used precisely this language for generalized Deutsch-type problems: trivalent functions of a trit were grouped into equipartitions, and the question was whether the corresponding functional classes could be separated by quantum state identifications [11]. In the present terminology, those examples emphasize that the useful immediate answer is a partition label, not a complete table of values.

This should not be read as a strict opposition between partial and full answers. Coarse-graining is a common way for a task to align with interference, but it is not the operative criterion. The operative criterion is structural compatibility: the relevant invariant must be supported by efficiently preparable inputs and oracle access, encoded by an efficiently implementable unitary, and extracted as an efficiently measurable, sufficiently probable final statistic. Arbitrary partitions of arbitrary black-box data need not have this property, and a structure-matched intermediate statistic may sometimes be classically processed into a complete answer.

Shor’s algorithm is a more sophisticated example. Modular exponentiation is computed reversibly, but the algorithm does not read all values of

$$f(x) = a^x \bmod N.$$

Instead, periodicity is sampled through a quantum Fourier transform. The success comes from algebraic structure: translation symmetry becomes spectral concentration. Shor’s algorithm is therefore an important warning against saying that interference is useful only for “partial” knowledge. The quantum subroutine produces samples from which the order can be inferred; with the standard random choice of a , checks, and repetitions, continued-fraction and greatest-common-divisor post-processing yields a nontrivial factor of N with high probability. The final classical answer may be complete even when the quantum readout is an invariant.

The same observation explains why generic predicates are not made easy by superposition. For an arbitrary Boolean string

$$b = (b_1, \dots, b_N),$$

the parity

$$b_1 \oplus b_2 \oplus \dots \oplus b_N$$

depends on every input bit. Exact quantum computation of parity requires exactly $\lceil N/2 \rceil$ queries, giving at most a factor-of-two improvement over the deterministic classical query complexity [12]. More generally, in the black-box query model, polynomial-method bounds show that for total Boolean functions the bounded-error quantum and classical query complexities are polynomially related, ruling out arbitrary exponential speedup for such functions [13]. Superposition alone is therefore not the resource. Structure-matched recombination is.

Appendix A gives a concrete interpolation of this point for the Bernstein–Vazirani problem. A unitary oracle is

modified so that a tunable part of the computational-basis path label is exported to an inaccessible witness register. The Boolean function is unchanged and the enlarged dynamics remains unitary, but the accessible interference pattern is progressively destroyed as more path information leaks. The appendix also shows that interference and full value recovery are composable: the surviving recombination recovers the coordinates still structurally available to it, and ordinary value queries can supply the rest.

Another illustration of this principle is provided by spatial-search and wave-communication constructions on regular lattices. The basic image is that of a coherent wave spread over a whole lattice. A locally marked site acts like a resonant defect: it is different enough from the background to be distinguished, but close enough in frequency to the extended lattice mode to exchange amplitude with it. The lattice then functions as a global carrier. In the lattice-search analysis of Hein and Tanner, this picture is made precise by an avoided crossing between the extended symmetric state and a localized defect state. The resulting dynamics rotates amplitude from the delocalized state into the marked localization. In their related wave-communication proposal, the same mechanism is extended to several marked vertices: with two marked defects, amplitude can be transferred between sender and receiver through the symmetric carrier state [14, 15].

This is a spectral, spatial-search analogue of Grover-type amplitude amplification: the advantage is not produced by simultaneously readable computational branches, but by coherent wave interference that converts a phase/scattering perturbation into an observable amplitude concentration. Grover search also clarifies why “structure” should not be inflated until it explains everything. The database is unstructured in the sense that it contains no exploitable character, period, or hidden-subgroup promise; the useful structure is only the two-dimensional geometry spanned by the uniform state and the marked subspace. The poverty of this structure is reflected in the quadratic, rather than exponential, speedup. The example also has a classical coherent-wave analogue. This supports the claim that the relevant recombination mechanism is wave-mechanical rather than intrinsically tied to measurement collapse or many-world branch readout. It is not, by itself, an efficiently scalable classical simulation: a wave realization pays for the full physical lattice or mode structure, whereas a quantum implementation may encode the corresponding walk space more economically when the structure permits. Interference in a medium is therefore a useful image for the positive mechanism, but not a sufficient account of computational advantage. The end-to-end economy depends on state preparation, oracle access, the encoding, and readout, typically in promise or oracle settings rather than for arbitrary total functions.

IV. GARBAGE IS NOT ERASED BY UNITARITY

A second misconception concerns garbage. Classical irreversible computation often hides the physical cost of deleting information. Quantum computation cannot do so, because closed-system evolution is unitary:

$$|\psi\rangle \mapsto U|\psi\rangle, \quad U^\dagger U = I.$$

A non-injective classical function cannot be implemented by a unitary acting only on the input register. It must be embedded reversibly, for example as

$$U_f : |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle.$$

The input, or an equivalent record, remains present.

If a reversible computation produces

$$|x\rangle |0\rangle |0\rangle \mapsto |x\rangle |f(x)\rangle |g(x)\rangle,$$

the garbage register $|g(x)\rangle$ cannot be erased by closed unitary evolution. It can be uncomputed; transferred to another register or an environment, thereby relocating the information; measured; or, when a physical memory is returned to a standard state for reuse, thermodynamically reset.

No isolated unitary erasure. Let $|\phi_1\rangle$ and $|\phi_2\rangle$ be two normalized pure states on a Hilbert space $\mathcal{H}$ which are not identical up to a global phase. Then there is no unitary U on $\mathcal{H}$ and no fixed normalized state $|0\rangle \in \mathcal{H}$ such that

$$U|\phi_1\rangle = U|\phi_2\rangle = |0\rangle.$$

Indeed, unitaries preserve inner products. If both equalities held, then

$$\langle\phi_1|\phi_2\rangle = \langle\phi_1|U^\dagger U|\phi_2\rangle = \langle 0|0\rangle = 1,$$

so $|\phi_1\rangle$ and $|\phi_2\rangle$ would coincide as rays, contradicting the assumption. This is the same elementary isometric obstruction that underlies the no-cloning and no-deletion results: closed reversible dynamics preserves distinguishability and therefore cannot freely merge, duplicate, or discard unknown alternatives.

Consequently, if $|g(x)\rangle_B$ is a garbage register produced by reversible evaluation of a non-constant function g on input x , there is no unitary acting on B alone that resets $|g(x)\rangle_B$ to $|0\rangle_B$ for all x . The garbage can instead be handled by (i) uncomputation, using the input register A , which is reversible and redistributes rather than destroys the information; (ii) coupling B to an ancillary system or environment, which transfers rather than destroys the information; if that receiving memory is later reset for cyclic use, the reset has a thermodynamic cost; or (iii) measurement of B , creating a classical record whose subsequent reset is subject to Landauer's principle.

Bennett's reversible-computation analysis shows how intermediate garbage can be uncomputed [16–18]; it does

not make physical erasure free. Landauer's principle locates the cost in resetting physically instantiated memory for cyclic use; unitary export or mere decoherence alone does not constitute erasure [19]. Whether the Landauer–Bennett account fully exorcises Maxwell's demon remains a point of philosophical dispute [20–22], but the operational lesson for quantum computation is unambiguous: unitary dynamics does not delete information.

V. COPYING AND DELETION ARE CONTEXTUAL

Classical information can be copied because it is encoded in distinguishable states. Quantum information cannot generally be copied. If a unitary cloner existed with

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \text{and} \quad U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle,$$

then unitarity preserves inner products: on the one side,

$$\langle\psi|\langle 0|U^\dagger U|\phi\rangle|0\rangle = \langle\psi|\phi\rangle\langle 0|0\rangle = \langle\psi|\phi\rangle,$$

while on the other side the cloned outputs give

$$\langle\psi\psi|\phi\phi\rangle = \langle\psi|\phi\rangle^2.$$

Equating these yields

$$\langle\psi|\phi\rangle(\langle\psi|\phi\rangle - 1) = 0,$$

so $|\psi\rangle$ and $|\phi\rangle$ are either orthogonal or coincide as rays. This is the no-cloning theorem [23, 24]. For mixed states, the no-broadcasting theorem says that a set of density operators can be broadcast if and only if they pairwise commute [25]. The no-deletion theorem gives the dual obstruction: an unknown quantum state cannot be deleted from one copy while leaving another intact [26].

These results identify the boundary of classicality. Within a fixed orthonormal basis one may copy basis labels,

$$|x\rangle|0\rangle \mapsto |x\rangle|x\rangle,$$

but this copies information only relative to that basis; it does not copy an arbitrary unknown state. In algebraic terms, classical copyability is confined to a commuting context.

Here a *context* means a maximal set of mutually commuting observables, equivalently a maximal Boolean subalgebra of the projection lattice. In the nondegenerate finite-dimensional case it corresponds to an orthonormal basis and the projectors generated by it.

Quantum computation therefore manipulates information that is not globally copyable or disposable. Only after a context is selected, typically by measurement or decoherence, do classical records emerge.

VI. CONTEXTUALITY AND THE ABSENCE OF A GLOBAL CLASSICAL HISTORY

Quantum algorithms move between incompatible descriptions: computational basis, phase basis, Fourier basis, stabilizer basis, and measurement basis. These basis changes already express complementarity. A state sharp in one basis is generally delocalized in another.

In Hilbert spaces of dimension at least three, the Kochen–Specker theorem strengthens this point: one cannot assign pre-existing values to all observables independently of the commuting context in which they are measured [27–29]. Operational versions extend contextuality to preparations, transformations, and unsharp measurements [30]. For single qubits the standard Kochen–Specker obstruction for projective measurements is absent; simple hidden-variable representations of projective qubit measurements can be constructed. This qualification is important: generalized measurements can exhibit Kochen–Specker-type contextuality even for a single qubit [31]. Nevertheless, complementarity already prevents joint sharp values for noncommuting projective observables and so blocks simultaneous classicalization of all measurement bases.

This matters for quantum computation, but the claim needs to remain narrow. When the relevant observables instantiate a Kochen–Specker scenario, there is no single context-independent assignment of pre-existing values across those contexts. This does not rule out every contextual hidden-variable description. An algorithm is a sequence of basis- and context-dependent unitary descriptions terminated by a chosen measurement context; what becomes classical is the final record, not a context-independent account of every intermediate observable.

This separates two questions that are often conflated. Structure-matched recombination explains where the accessible answer comes from; resource-theoretic obstructions explain why the same transformation is not, in general, reproducible by an efficient classical stochastic process. Neither observation is sufficient on its own, and contextuality is not asserted here to be a necessary-and-sufficient explanation of every quantum advantage.

Resource-theoretic results support this perspective, but with important qualifications. Stabilizer operations admit efficient classical simulation by the Gottesman–Knill theorem [32, 33]; hence universal quantum advantage in the circuit model requires resources outside the efficiently simulable stabilizer fragment. Depending on the formalism and dimension, these resources may be described operationally as magic states, phase-space theoretically as Wigner-function negativity, or foundationally as contextuality [34–36]. These characterizations are related but not identical. In particular, the clean necessity results connecting contextuality with magic-state quantum computation are most direct in odd-prime-dimensional qudit settings, while the qubit case involves additional subtleties. The stable message is therefore not that contextuality alone is the complete source of

speedup. Rather, quantum algorithms exploit transformations between incompatible contexts, and genuine advantage requires resources that prevent the whole computation from being represented as a single efficiently simulable classical stochastic process.

VII. QUANTUM UNIVERSALITY VERSUS TURING UNIVERSALITY

The word “universal” is used in two substantially different senses in computation theory and quantum information theory. Confusing them leads to conceptual errors.

In the quantum-circuit literature, a finite gate set is called universal if circuits over that set can approximate arbitrary unitary transformations for each fixed number n of qubits, typically to arbitrary accuracy in operator norm. In this sense, universality concerns dense generation of groups such as $U(2^n)$ or $SU(2^n)$ and hence finite-dimensional controllability. Density is an approximation statement; efficient compilation is an additional algorithmic requirement.

Turing universality is different. A Turing-universal device acts on finite symbolic inputs of unbounded length and, with appropriate finite encodings, can simulate arbitrary effective procedures [37, 38]. This is the setting in which partial recursive functions, undecidability, universal simulation, and the halting problem arise.

Thus a finite quantum circuit, by itself, does not define the same kind of object as a Turing machine. A circuit on n qubits implements a unitary transformation

$$U_n : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^{2^n}.$$

It is a finite-dimensional, norm-preserving map. It has fixed width and depth, no unbounded tape, no open-ended input parsing, and no intrinsic recursion. It also has no autonomous mechanism for extending its own computation to arbitrary symbolic length.

A family of circuits $\{C_n\}_{n \geq 1}$ may compute a language or function in the complexity-theoretic sense only when an effective uniform rule generates the circuit C_n from n . That uniformity condition is itself algorithmic and may be supplied classically or by a specified higher-level quantum model such as a quantum Turing machine [7, 39, 40]. Thus quantum circuit families and quantum Turing machines are not identified merely by the existence of a universal gate set; their comparison requires explicit uniformity and simulation assumptions.

This distinction is important for the phrase “quantum computable function.” In the finite-circuit model, a circuit computes only a finite input-output relation, usually probabilistically and after measurement. A uniform circuit family computes a sequence of such relations. A quantum Turing machine or another uniform quantum model can be compared to classical Turing computation, but then the usual Church–Turing issue concerns computability, while the quantum advantage concerns complexity. Under the usual effectiveness assumptions—in

particular, uniform circuit generation and computable gate entries—standard quantum computation does not enlarge the class of computable classical functions beyond the partial recursive ones; rather, it may reduce the resources needed for some computable tasks. Allowing noncomputable gate parameters would change the model by embedding uncomputable information.

A related point concerns halting. Classical irreversible computation may enter an absorbing terminal configuration: many distinct pre-halting configurations can evolve into one stable halt state. Such many-to-one absorption is not a primitive of closed unitary dynamics. If $U|h\rangle = |h\rangle$ and $U|\psi\rangle = |h\rangle$, then injectivity of U gives $|\psi\rangle = |h\rangle$. Thus a closed unitary may coherently mark, clock, or correlate halting information, but it cannot collapse distinct computational histories into a single absorbing halt state without exporting information to additional degrees of freedom.

This elementary observation is not a no-go theorem against quantum Turing machines. Standard quantum models implement operational termination through halt flags, clocks, external control, or measurement protocols [7, 39, 41, 42]. The point is only that the classical absorbing-state picture is not itself a closed-unitary primitive.

This reinforces the main thesis. Quantum circuit universality is a powerful notion of finite-dimensional controllability and approximation. Turing universality is a notion of symbolic, unbounded, self-referential computation under an explicit uniformity condition. They intersect in uniform models of quantum computation, but they are not identical. Treating “universal quantum computation” as if it automatically meant Turing universality obscures the distinction between unitary dynamics, measurement, and algorithmic control.

VIII. CONCLUSION

Quantum speedup, in the sense considered here, involves a conjunction. Structure-matched recombination explains where the accessible answer comes from: problem structure is encoded in coherent amplitudes and concentrated in a measurement context by interference. Resource constraints explain why the same transformation is not, in general, reproducible by an efficient classical stochastic process. Neither condition alone suffices.

Quantum computation is therefore not classical computation performed in exponentially many parallel branches whose results can later be inspected. Its power lies in reversible amplitude manipulation and phase-sensitive computation, with output extracted in a chosen measurement context. An advantage requires an end-to-end efficient encoding, transformation, and readout; structural matching is a useful explanatory criterion, not a general sufficient condition. Often the immediately measured object is a partition label, period, parity, or marked subspace rather than a full classical description

of the instance. But such coarse-graining is a common consequence of structure-matched recombination, not the criterion itself; an invariant may still determine a complete classical answer after post-processing.

Exponential Hilbert-space dimension should not be mistaken for exponentially accessible classical storage. Large dimension provides room for structured embeddings and interference, but readout is constrained by measurement, Holevo bounds, and the chosen context. In this sense high-dimensional quantum state space is a resource for controlled, structure-matched recombination, not for unrestricted branchwise inspection.

The negative constraints are just as important. Garbage cannot be erased by unitary dynamics; it must be uncomputed or relocated, and resetting a physical memory for reuse is thermodynamic. Unknown quantum states cannot be freely copied, broadcast, or deleted. Incompatible contexts—and, where a Kochen–Specker scenario is involved, contextuality—preclude a single context-independent assignment of pre-existing values across all the relevant observables.

Nor should circuit or unitary universality be confused with Turing universality. A finite universal gate set provides dense control over unitary transformations; it is not, by itself, a symbolic machine with arbitrary parsing, recursion, self-reference, and a classical halting semantics. In closed unitary dynamics there is no nontrivial absorbing halt state of the classical many-to-one kind. Termination and output extraction require clocks, flags, measurements, open-system records, or external control.

Thus quantum computation is neither magic parallelism nor ordinary classical computation in disguise. It is context-dependent, thermodynamically embedded, reversible information processing whose useful answers must be engineered as interference phenomena. More pointedly, quantum speedup is interference-enabled, structure-matched amplitude recombination; coarse-graining the target predicate is one common way to obtain the required match, but it is the match, not the coarseness, that does the work.

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Appendix A: A unitary leakage knob for Bernstein–Vazirani interference

The Bernstein–Vazirani algorithm gives a particularly transparent example of quantum speedup by interference rather than by any literal parallel readout of classical function values [6, 39, 43]. Let $a \in \mathbb{F}_2^n$ be an unknown string and define

$$f_a(x) = a \cdot x \pmod{2}, \quad x \in \mathbb{F}_2^n. \quad (\text{A1})$$

The usual reversible oracle is

$$O_a |x\rangle |y\rangle = |x\rangle |y \oplus f_a(x)\rangle. \quad (\text{A2})$$

With the target register initialized in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, this oracle acts by phase kickback,

$$O_a |x\rangle |-\rangle = (-1)^{a \cdot x} |x\rangle |-\rangle.$$

Thus

$$\begin{aligned} |0^n\rangle |-\rangle &\xrightarrow{H^{\otimes n}} 2^{-n/2} \sum_x |x\rangle |-\rangle \\ &\xrightarrow{O_a} 2^{-n/2} \sum_x (-1)^{a \cdot x} |x\rangle |-\rangle \\ &\xrightarrow{H^{\otimes n}} |a\rangle |-\rangle. \end{aligned} \quad (\text{A3})$$

Equivalently, the final amplitude of $|z\rangle$ is

$$2^{-n} \sum_{x \in \mathbb{F}_2^n} (-1)^{(a \oplus z) \cdot x} = \delta_{z,a}. \quad (\text{A4})$$

The hidden string is therefore recovered exactly with one coherent query. The essential step is not the evaluation of many classical values, but the conversion of a global phase character into a computational-basis population by a Fourier transform over $\mathbb{F}_2^n$.

We now introduce a reversible modification that degrades this interference without changing the Boolean function f_a . Let $S \subseteq \{1, \dots, n\}$ have cardinality k , and let $\bar{S}$ denote its complement. For $x \in \mathbb{F}_2^n$, write

$$x_S = (x_i)_{i \in S} \in \mathbb{F}_2^k, \quad x_{\bar{S}} = (x_i)_{i \in \bar{S}} \in \mathbb{F}_2^{n-k},$$

where a fixed ordering of the coordinates in S and $\bar{S}$ is understood. Thus x_S and $x_{\bar{S}}$ denote the subvectors of x supported on S and on its complement, respectively. Define the unitary permutation

$$W_a^S |x\rangle |y\rangle |e\rangle = |x\rangle |y \oplus f_a(x)\rangle |e \oplus x_S\rangle, \quad (\text{A5})$$

where the third register is a witness, garbage, or environmental register. The oracle still computes the same function f_a , and the enlarged map is reversible. However, if the third register is not coherently available to the algorithm, then it carries which-path information about the computational-basis label x . Here “which-path” is meant in the standard interferometric sense: a record relative

to a chosen computational basis, not an independently readable classical computation carried by a component of the superposition. In this respect the construction is a tunable version of the familiar which-way suppression of interference [44–47]. It is also closely related to Aaronson’s discussion of garbage in quantum query models, where sufficiently informative garbage in an oracle response is suggested to behave, in appropriate settings, like decoherence or measurement of the response [48].

Proposition: partial loss of the Bernstein–Vazirani interference pattern. Let the query register be initialized in $|+\rangle^{\otimes n}$, the answer register in $|-\rangle$, and the witness register in $|0^k\rangle$. Apply W_a^S , trace out the witness register, and then apply $H^{\otimes n}$ to the query register. The final measurement distribution on the query register is

$$\Pr(z) = 2^{-k} \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}]. \quad (\text{A6})$$

Thus one coherent Bernstein–Vazirani-style query determines the $n - k$ components of a outside S with certainty, while the observed coordinates z_S are uniformly distributed over $\mathbb{F}_2^k$ and carry no information about a_S . In particular, the usual one-query speedup survives only on those coordinates whose computational-basis path labels have not been exported to the inaccessible witness.

Operationally, after this first query one may recover the remaining coordinates by k ordinary value queries on the standard basis vectors e_i , $i \in S$, since $f_a(e_i) = a_i$. This provides an explicit completion strategy with query count $1 + k$ for $k < n$. At the endpoint $k = n$, the first Bernstein–Vazirani-style query gives the uniform distribution and is best omitted, so the standard n value queries suffice. Thus the interference step and ordinary value recovery are not opposed: a structure-matched interference readout can supply the part of the answer that remains coherently recombinable, while additional value queries can complete the full classical string. Hence this simple strategy interpolates as

$$Q_{\text{strat}}(n, k) = \begin{cases} 1 + k, & 0 \leq k < n, \\ n, & k = n. \end{cases} \quad (\text{A7})$$

Equation (A7) describes this explicit completion strategy, not a general optimality theorem for arbitrary adaptive use of the enlarged leaked oracle. The structural fact proved below is narrower and sufficient for the present purpose: the accessible phase-interference channel contains no dependence on a_S . Within the stated value-query completion, determining the remaining k -bit string a_S requires k independent linear equations.

Proof. After the first Hadamard transform and the application of W_a^S , the joint state is

$$|\psi_a^S\rangle = 2^{-n/2} \sum_{x \in \mathbb{F}_2^n} (-1)^{a \cdot x} |x\rangle |-\rangle |x_S\rangle. \quad (\text{A8})$$

Tracing out the witness register gives the reduced density operator on the query register

$$\rho_a^S = 2^{-n} \sum_{\substack{x, x' \in \mathbb{F}_2^n \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x')} |x\rangle \langle x'|. \quad (\text{A9})$$

Hence coherences remain only between computational-basis components that agree on the coordinates in S . The witness register has therefore implemented a perfect which-path record for those coordinates.

Now apply $H^{\otimes n}$ and measure in the computational basis. For $z \in \mathbb{F}_2^n$,

$$\Pr(z) = \langle z | H^{\otimes n} \rho_a^S H^{\otimes n} | z \rangle \quad (\text{A10})$$

$$= 2^{-2n} \sum_{\substack{x, x' \in \mathbb{F}_2^n \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x') + z \cdot (x \oplus x')}. \quad (\text{A11})$$

Set $t = x \oplus x'$. The constraint $x_S = x'_S$ is equivalent to $t_S = 0$. For each such t , there are 2^n choices of x . Therefore

$$\Pr(z) = 2^{-n} \sum_{\substack{t \in \mathbb{F}_2^n \\ t_S = 0}} (-1)^{(a \oplus z) \cdot t} \quad (\text{A12})$$

$$= 2^{-n} \sum_{t_S \in \mathbb{F}_2^{n-k}} (-1)^{(a_S \oplus z_S) \cdot t_S}. \quad (\text{A13})$$

By character orthogonality over $\mathbb{F}_2^{n-k}$,

$$\sum_{t_S \in \mathbb{F}_2^{n-k}} (-1)^{(a_S \oplus z_S) \cdot t_S} = \begin{cases} 2^{n-k}, & z_S = a_S, \\ 0, & z_S \neq a_S. \end{cases} \quad (\text{A14})$$

This proves Eq. (A6).

It remains to identify what information remains available through the interference channel. Consider the effective reduced oracle obtained from one call to W_a^S followed by tracing out the witness register. For the phase-kickback use of the answer register, this reduced channel on the query register is

$$\mathcal{E}_a^S(\rho) = D_a \left(\sum_{s \in \mathbb{F}_2^k} P_s \rho P_s \right) D_a^\dagger, \quad (\text{A15})$$

where

$$D_a = \sum_x (-1)^{a \cdot x} |x\rangle \langle x| \quad (\text{A16})$$

and

$$P_s = \sum_{x: x_S = s} |x\rangle \langle x|. \quad (\text{A17})$$

Equivalently, in matrix elements,

$$\mathcal{E}_a^S(\rho) = \sum_{\substack{x, x' \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x')} \rho_{xx'} |x\rangle \langle x'|. \quad (\text{A18})$$

Inside each block $x_S = x'_S = s$, the contribution of a_S cancels:

$$a \cdot (x \oplus x') = a_S \cdot (x_S \oplus x'_S). \quad (\text{A19})$$

Therefore the reduced phase-interference channel depends on a only through a_S . The hidden components a_S are not merely hard to extract by the final Hadamard transform; they are absent from the accessible phase channel once the witness has been traced out.

The same dephasing in the x_S basis occurs for any preparation of the answer register, since the witness records x_S independently of the answer-register state. The dephasing removes the dependence on a_S from the accessible *phase-interference* channel; it does not remove ordinary value information carried by the answer register. Consequently, once a_S has been obtained by the surviving Bernstein–Vazirani interference, the remaining components a_S can be recovered by standard value-query use of

$$f_a(x) = a_S \cdot x_S + a_{\bar{S}} \cdot x_{\bar{S}}. \quad (\text{A20})$$

Since a_S is then known, a standard value query supplies one linear equation over the unknown vector $a_S \in \mathbb{F}_2^k$. Querying the standard basis vectors e_i , $i \in S$, recovers these k components directly. This proves the stated interpolation strategy in Eq. (A7). $\square$

The two extremes make the physical point explicit. For $k = 0$, no computational-basis path label is exported and the ordinary one-query Bernstein–Vazirani interference pattern is recovered. For $k = n$, the witness records the entire computational-basis label x , all off-diagonal terms in that basis are removed from the algorithm’s accessible state, and the final Hadamard transform has no global phase pattern left to recombine. The Boolean function f_a has still been evaluated coherently by a unitary permutation on the enlarged system. What has been lost is not evaluation but recombination.

This example shows that reversibility alone is not the relevant criterion. The map W_a^S is a unitary permutation on the enlarged Hilbert space. If the witness register remains coherently accessible, one may uncompute it and restore the full interference pattern. If, however, the witness register has escaped into an inaccessible environment, then the reduced dynamics available to the algorithm is no longer reversible. The witness register is precisely the uncomputed garbage identified in the main text as an obstruction to free unitary erasure, here promoted to a tunable knob. The one-query Bernstein–Vazirani readout advantage is lost in this leakage model not because the Boolean function f_a has changed, nor because the global dynamics has ceased to be unitary, but because an inaccessible which-path record selects a computational-basis context incompatible with the later Fourier-basis recombination.

Pointedly stated, this demonstrates that a quantum computer does not “compute all branches at once and let us independently read them”—where the scare-quoted

branch language names precisely the classical picture being rejected. If it did, leaking the computational-basis path labels to a witness register would not matter: the function $f_a(x)$ was still computed. Instead, the appendix shows that the speedup relies on *recombining amplitudes*

by interference in the basis matched to the hidden linear character. When uncomputation is blocked by an inaccessible witness, the accessible dynamics is dephased in the computational basis, the interference is quenched, and the lost coordinates must be recovered, if at all, by ordinary value queries.

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